WMCS045-05 2026
Lab Assignment #7
Deadline: 2026-10-21 **Group number: _____**

1. Data Structure Optimization

Consider the following code:

```
#include <iostream>
#include <algorithm>
#include <vector>
#include <cstring>

using namespace std;

const int n = 16384;

struct Small {
    char flg1;
    double x[7];
    char flg2;
};

struct Big {
    int rank;
    double x[n];
};

bool sortSmallByx0(const Small &a, const Small &b) { return a.x[0] < b.x[0]; }
bool sortBigByRank(const Big &a, const Big &b) { return a.rank < b.rank; }

int main()
{
    vector<Small> vsmall(n);
    vector<Big> vbig(n);

    for (int i; i < n; i++)
        vsmall[i].x[0] = n - i;

    for (int i; i < n; i++) {
        vbig[i].rank = n - i;
        std::memset(vbig[i].x, 0, sizeof(vbig[i].x));
    }

    sort(vsmall.begin(), vsmall.end(), sortSmallByx0);

    sort(vbig.begin(), vbig.end(), sortBigByRank);

    return 0;
}
```

In lecture #12 the following data structure optimization techniques have been presented: data packing, data padding, field re-ordering, and structure splitting.

Address the following tasks:

a) Explain for each of the techniques whether they could be used for optimizing the data structures `Small` and `Big`.

b) Choose and implement an optimization of `Small` and `Big`.

c) Use perf or PAPI to provide empirical evidence of the performance improvement and changes. Provide results for different performance counters and try to explain the observations.

You are free to change the benchmark and consider the different data structures separately.

2. MPI Point-to-Point Communication Benchmarking

Consider a communication benchmark using MPI that fulfils the following requirements:

1. The number of MPI ranks N is provided as an argument to `mpiexec`
2. The process with rank i sends a message to the process with rank $(i + 1) \bmod N$
3. The process with rank i receives a message from the process with rank $(i-1) \bmod N$
4. The message size is configurable
5. The benchmark reports the communication bandwidth, which is determined using `MPI_Wtime()`. It is recommended to repeat the communication M times to obtain reliable measurements.

Address the following tasks:

a) Implement the earlier described benchmark

b) Measure the bandwidth for messages lengths in the range of 128 byte to 4096 byte using
   - Two processes running on the same node
     (Slurm options: `--nodes=1 --ntasks-per-node=2`)
   - Two processes running on different nodes
     (Slurm options: `--nodes=2 --ntasks-per-node=1`)

c) Bonus task: Compare results using the nodes in partition `cn-ib` and `cn-eth`